# JASWANT JAYACUMAAR

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## Professional Summary

Data Analyst with 3+ years across e-commerce, consulting, and scientific research, specialising in Python, SQL/BigQuery, and building full-stack analytics platforms end-to-end. Currently own the contribution-margin reporting platform at Printerpix and built an autonomous LLM-driven bidding agent with human-approval controls, recommending bids across ~$10M of annual Google Ads spend in 9 international markets, integrating LLM-powered systems (LangGraph, Gemini, Claude) throughout the analytics stack. Strong statistical foundation (Bayesian inference, MCMC, and machine learning) from postgraduate astrophysics research. Proven record of turning messy multi-source data into automated pipelines, dashboards, and AI tools that commercial teams use daily.

## Skills

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| **Programming:** | Python (pandas, NumPy, scikit-learn, LightGBM, TensorFlow), SQL (BigQuery, PostgreSQL), TypeScript |
| **Data Engg. & Cloud:** | BigQuery (scheduled queries, partitioning, cost optimisation), Google Cloud (Cloud Run, Cloud Scheduler, Cloud Build), PostgreSQL, Supabase (RLS, RBAC), ETL pipelines, data-warehouse migration & reconciliation, data-quality monitoring & alerting |
| **AI & Agentic Systems:** | LangGraph multi-agent pipelines, LLM judging chains (critic/arbiter), human-in-the-loop approval design, RAG, LLM evaluation & back-testing, Gemini/Claude/GPT APIs, MCP-accelerated workflows, Claude Code, Cursor |
| **Statistics & ML:** | Bayesian inference, MCMC, statistical change detection (z-score gating), conversion-maturation & time-series modelling, regression, Monte Carlo simulation, predictive modelling, back-testing & validation |
| **Marketing Analytics:** | Google Ads API & auction insights, Bing Ads API, GA4, Search Console, Merchant Center, GTM & server-side tracking, SEM bid automation, ROAS/CM optimisation, competitor-price intelligence |
| **Web, BI & Delivery:** | Next.js/React, FastAPI, Flask, Express, Docker, Vercel, Railway, Power BI (DAX), Looker Studio, Git/GitHub, pytest/TDD |

## Experience

**PrinterPix UK Ltd** — London, United Kingdom
*Data Analyst, e-Commerce* — Mar 2026 – Present

- Designed and built an autonomous nightly bidding agent and its human-approval AI dashboard, covering Search, Shopping, Broad and PMax campaigns across 9 markets and 17 Google Ads accounts: statistical change detection (calibrated z-score gates, back-tested against 232 historical target changes) feeds a multi-model LLM judging chain (decider, blind critic, arbiter, evidence check) proposing bid, budget, negative-keyword and search-term moves, written directly to Google Ads on approval with rollback and full audit trail
- The agent recommends bids across ~$10M of annual SEM spend behind a human approval gate, applying 10,000+ approved campaign changes and classifying 27,000+ search terms per month, eliminating 25+ hours/week of manual bid management, with a nightly outcomes loop that measures every decision against settled results to continuously improve its recommendations
- Solely designed, built, and own the company's Contribution Margin (CM) reporting platform: a full-stack web application (Next.js, Express, BigQuery, Supabase; deployed on Vercel/Railway) replacing legacy Looker Studio dashboards with 11 analytical pages of live daily and hourly profitability, marketing-cost, SEM, auction-insight, conversion-maturation and AI-bidding-decision analytics across 9 international markets
- Built a same-day contribution-margin prediction system: models immature Day-1 Google Ads data through measured conversion-maturation and tracking-loss curves to forecast settled CM per campaign and ad group across Search, Shopping and PMax, backed by nightly BigQuery materialisers, a validation view tracking predicted against settled CM (typically within 5-7%), and query optimisation that cut filter scans from 51.5 GB to 1.2 MB (41,000x cheaper)
- Led the migration of the warehouse ingestion layer from MS SQL Server to PostgreSQL: replaced legacy cron feeds with a PostgreSQL-to-BigQuery sync pipeline, backfilled order history to 2018 (6.6M+ rows) within BigQuery load quotas via a metered backfill driver, repointed 29 scheduled queries through a phased, reconciliation-gated cutover

(including one executed rollback), and root-caused three legacy data defects, one of which had been overstating contribution margin for months; now protected by automated health-check alerting

- Deployed a fleet of scheduled AI monitoring agents across the paid-media estate (account audits, auction-position tracking, search-term mining, feed and landing-page checks) and led a seven-market account-hygiene overhaul that removed 100,000+ conflicting negative keywords with full change registers and rollback files
- Took ownership of the company's end-to-end pricing-intelligence pipeline: automated competitor-price scraping (9 regions × 10 product lines), BigQuery consolidation, and a regression + Monte Carlo price-optimisation engine feeding weekly price recommendations into the SEM dashboard
- Engineered and documented the BigQuery warehouse powering all of the above: scheduled queries consolidating Google Ads, GA4, Bing Ads, Merchant Center, Search Console, Klaviyo, and competitor-pricing data, with full pipeline DAGs, source mappings, and calculation references authored in Confluence

**TELUS Digital AI Data Solutions** London, United Kingdom
*Data Analyst (Freelance)* Oct 2025 – Feb 2026

- Evaluated and validated geospatial, search and natural-language (NLP) datasets for a leading global mapping platform, applying detailed research guidelines to assess accuracy, relevance and completeness, maintaining high QA audit scores on rubric-based quality checks

**3i Infotech UK** London, United Kingdom
*Associate (Process Mining Analyst)* Nov 2023 – Jul 2025

- Conducted EDA on business and financial datasets using Python and SQL, cleaning and integrating data from multiple sources, delivering insights that improved order processing time by 15%
- Built and maintained reports in Excel and Power BI, including interactive dashboards with DAX measures, data modelling and relationships to visualise KPIs, enabling data-driven decisions and commercial insight for sales teams
- Provided stakeholder support through cross-functional collaboration and presentations, translating complex statistical findings into actionable stories under tight deadlines

## Projects

**Timing Pulsars with the Thousand Pulsar Array on MeerKAT** Manchester, United Kingdom
*Jodrell Bank Centre for Astrophysics, University of Manchester* Sept 2021 – Nov 2023

- Produced updated timing solutions (ephemerides) for 179 pulsars with TEMPO2, building Python automation around the timing pipeline to batch-process and clean irregularly sampled time-series data from the Lovell and MeerKAT radio telescopes in a Linux environment
- Measured spin-down-rate variations for 8 pulsars by benchmarking two independent methods across three datasets: Bayesian Gaussian-process red-noise modelling with ensemble MCMC (run_enterprise, emcee, 32 chains) versus harmonic-whitening (fitwaves), quantifying uncertainties and demonstrating the Bayesian approach's robustness to irregularly sampled data
- Integrated observations from two telescopes into a combined dataset, handling cross-instrument clock offsets, and generated publication-quality figures and summary statistics (Matplotlib) for the dissertation

## Certifications

**KPMG Data Analytics Consulting Virtual Internship Program** 2023

- Assessed data quality and completeness in preparation for data analysis to identify high-value customers based on customer demographics and presented insights using data visualizations

**Cognizant Artificial Intelligence Virtual Experience Program** 2023

- Performed exploratory data analysis on sales and sensor data, built a machine learning model and implemented algorithm production to predict stock levels of products, and reviewed algorithm performance

## Education

**The University of Manchester** Manchester, United Kingdom
*Master of Science by Research in Astronomy and Astrophysics* 2021 – 2023

**Vellore Institute of Technology** Chennai, India
*Bachelor of Technology in Mechanical Engineering, 1st Class with Distinction* 2016 – 2020